

Survey 1

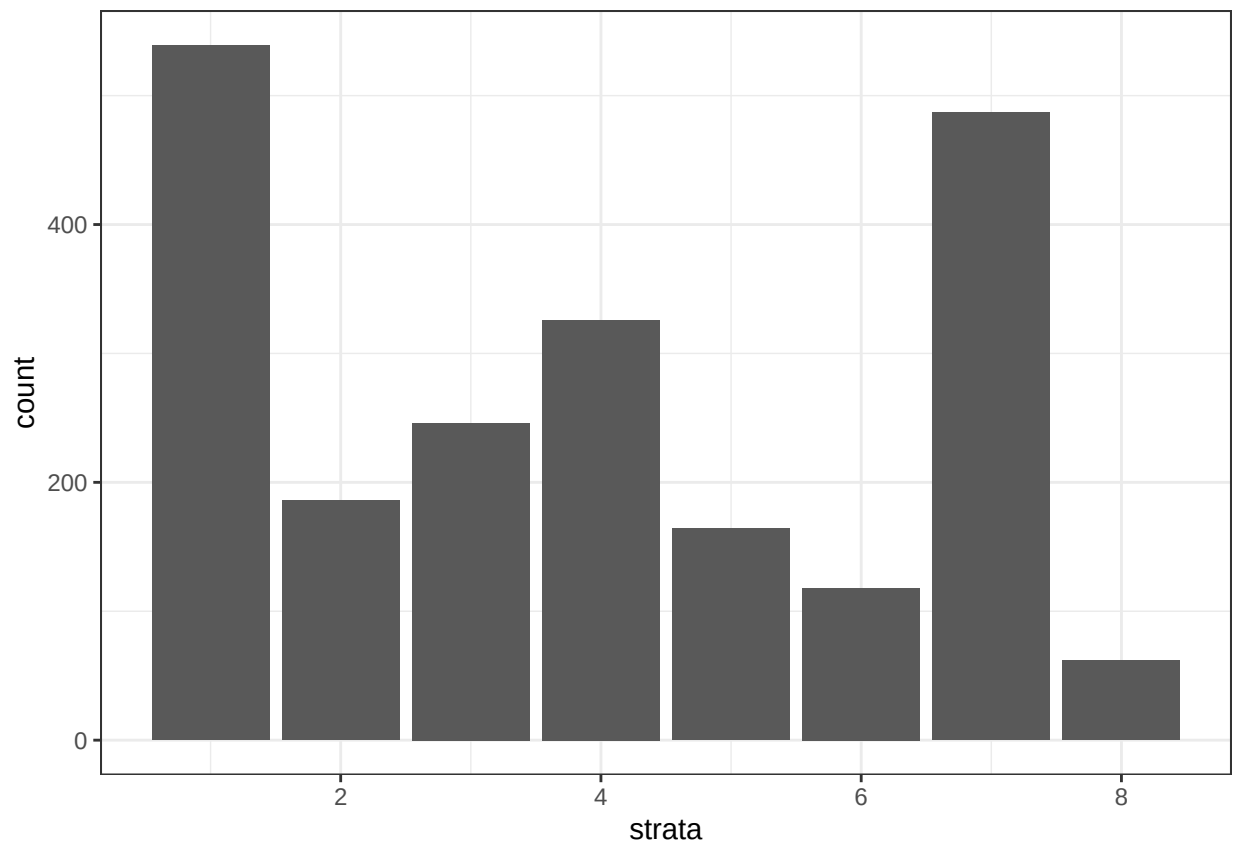
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```
library(tidyverse)
theme_set(theme_bw())
```

Use the following simulated data.

```
set.seed(2)
meanVals<-rlnorm(8,log(10),0.5)
habArea<-trunc(rlnorm(8,log(200),0.5))
area<-data.frame(strata=1:8) %>%
  mutate(habArea=habArea)%>%
  uncount(habArea)%>%
  mutate(N=rpois(sum(habArea),meanVals[strata]))
ggplot(area,aes(x=strata))+
  geom_bar()
```



1. Take a sample of size 50 and calculate the total numbers of animals, it's standard error, and 95% confidence interval using simple random sampling

```
sum(area$N)
```

```
## [1] 23163
```

```
n<-50
dataSimple<-sample(1:nrow(area),size=n,replace=FALSE)
simple<-area[dataSimple,] %>%
  summarize(meanCount=mean(N),
             seCount=sd(N)/sqrt(n()),
             n=n(),
             N=meanCount*nrow(area),
             seN=seCount*nrow(area),
             lci=N-2*seN,
             uci=N+2*seN)
simple %>% mutate(across(where(is.numeric),round,2))
```

```
## Warning: There was 1 warning in 'mutate()'.
## i In argument: 'across(where(is.numeric), round, 2)'.
## Caused by warning:
## ! The '...' argument of 'across()' is deprecated as of dplyr 1.1.0.
## Supply arguments directly to '.fns' through an anonymous function instead.
##
## # Previously
## across(a:b, mean, na.rm = TRUE)
##
## # Now
## across(a:b, \(x) mean(x, na.rm = TRUE))
##
##   meanCount seCount  n      N    seN    lci    uci
## 1      12.2    0.87 50 25961.6 1851.16 22259.28 29663.92
```

Also take a stratified sample of the same sample size and calculate the total numbers, se and CI. Is simple random or stratified more precise?

```
SamplesPerStrata<-round(n*meanVals/sum(meanVals)) #variance=mean for Poisson
SamplesPerStrata
```

```
## [1] 4 6 12 3 5 6 8 5
```

```
strataSample<-NULL
for(i in 1:8) strataSample<-
  c(strataSample,sample(which(area$strata==i),
                        size=SamplesPerStrata[i],
                        replace=FALSE))
table(area$strata[strataSample])
```

```
##
## 1 2 3 4 5 6 7 8
## 4 6 12 3 5 6 8 5
```

```

stratified<- area[strataSample,] %>%
  group_by(strata)%>%
  summarize(meanCount=mean(N),
            seCount=sd(N)/sqrt(n()),
            n=n()) %>%
  mutate(area=habArea,
         N=meanCount*area,
         seN=seCount*area) %>%
  ungroup()

stratified %>% mutate(across(where(is.numeric),round,2))

```

```

## # A tibble: 8 x 7
##   strata meanCount seCount    n area    N    seN
##   <dbl>    <dbl>   <dbl> <dbl> <dbl> <dbl> <dbl>
## 1     1        3.5    1.32     4  539 1886.  713.
## 2     2       10.7    1.26     6  186 1984  234.
## 3     3       21.2    1.54    12  246 5207  379.
## 4     4        5.67   0.33     3  326 1847.  109.
## 5     5        8.6    1.21     5  164 1410.  198.
## 6     6        9.67   1.63     6  118 1141.  192.
## 7     7       12.2    1.26     8  487 5966.  616.
## 8     8       11.4    1.08     5   62  707.  66.8

```

```

stratTotal<-stratified %>% summarize(N=sum(N),
                                   seN=sqrt(sum(seN^2)),
                                   lci=N-2*seN,
                                   uci=N+2*seN)
stratTotal %>% mutate(across(where(is.numeric),round,2))

```

```

## # A tibble: 1 x 4
##       N    seN    lci    uci
##   <dbl> <dbl> <dbl> <dbl>
## 1 20148. 1085. 17978. 22319.

```

Make plot

```

bind_rows(list(simple=simple,
               stratified=stratTotal),
  .id="Method")%>%
  ggplot(aes(x=Method,y=N,ymin=lci,ymax=uci))+
  geom_point()+
  geom_errorbar(width=0.1)+
  geom_hline(yintercept=sum(area$N),color="red")

```

